

GLUCOSE



INSULIN



SUPER MARIO DIACARE

Ashika Ramesh | Major Project | Sem 1 2024



MD
IT



RMIT
UNIVERSITY

Master of Design, Innovation and Technology

Major Project - Semester 1 2024

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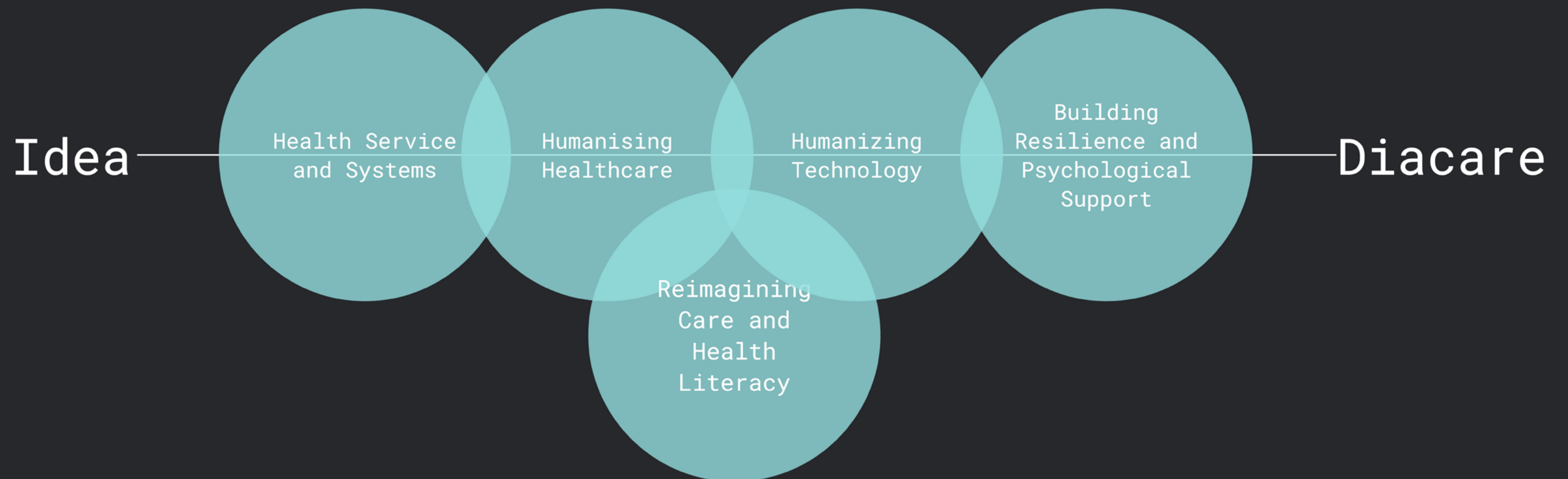
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ABSTRACT

As the global prevalence of Type 2 diabetes continues to rise, particularly among children, managing the overwhelming amount of health information has become challenging. Traditional textual materials often fail to engage children, leading to poor self-care practices. This project explores the use of interactive computer games to improve health literacy and diabetes management.

Utilizing an open-source version of the Mario Brothers game, I intended to incorporate educational material to create an engaging learning environment.

This design research seeks to highlight the possibilities for gamifying Type 2 Diabetes education to empower children and Teenagers.

Objectives:

Enhance Internal Health Locus of Control (HLOC): Strengthen the belief among individuals that they can influence their health outcomes through proactive self-management.

Improve Health Literacy: Deliver easy-to-understand health information to facilitate informed decision-making and improve adherence to self-care routines.

Integration of Technology: By leveraging interactive gaming, this project aims to bridge the gap in existing educational methods, offering a dynamic platform that enables children to understand and manage their health effectively.

KEYWORDS

Quality of life (QOL); Health Literacy; Health locus of control points (HLOC); Type 2 Diabetes (T2D); Game Design ; Education

Introduction

We live in an information-rich world that is always fighting for our attention. Current technology is rapid and insufficient, and it just serves to assist us forget about our current situation. The increase in healthcare information, both scientific and popular, is causing information overload for all actors in the healthcare system.

*"Information overload is associated with higher levels of stress and poor self-care performance."
- Wang and Voss 2022*

The goal of this research is to educate children and teenagers about Type 2 diabetes. Type 2 diabetes, the most prevalent type of diabetes, is a disease that happens when your blood glucose, commonly known as blood sugar, becomes too high. Blood glucose is your primary source of energy, and it is obtained from the foods we eat.

The good news is that you can make efforts to avoid or slow the progression of type 2 diabetes. Quality of life, according to the World Health Organization (WHO), is an individual's view of their place in life within the context of the culture and value systems in which they live, as well as their objectives, expectations, standards, and worries. The challenge for the 536.6 million individuals globally with diabetes, a figure that is projected to rise to 783.2 million by 2045—is not simply managing a chronic condition but also improving life quality across various areas.

The relationship between psychological well-being, self-management, and societal participation becomes critical in Type 2 diabetes management. Self-care, an essential component of everyday diabetes control, includes eating healthy foods, being physically active, taking medications correctly, and periodically testing blood sugar levels. An individual's health locus of control (HLOC), a psychological framework that helps predict self-management behaviors, has a significant impact on their self-care practice.

However, effective self-care is also dependent on health literacy—an individual's capacity to receive, process, and comprehend basic health information required to make proper health decisions. Low health literacy can impair disease management, resulting in poorer health outcomes and a lower quality of life.

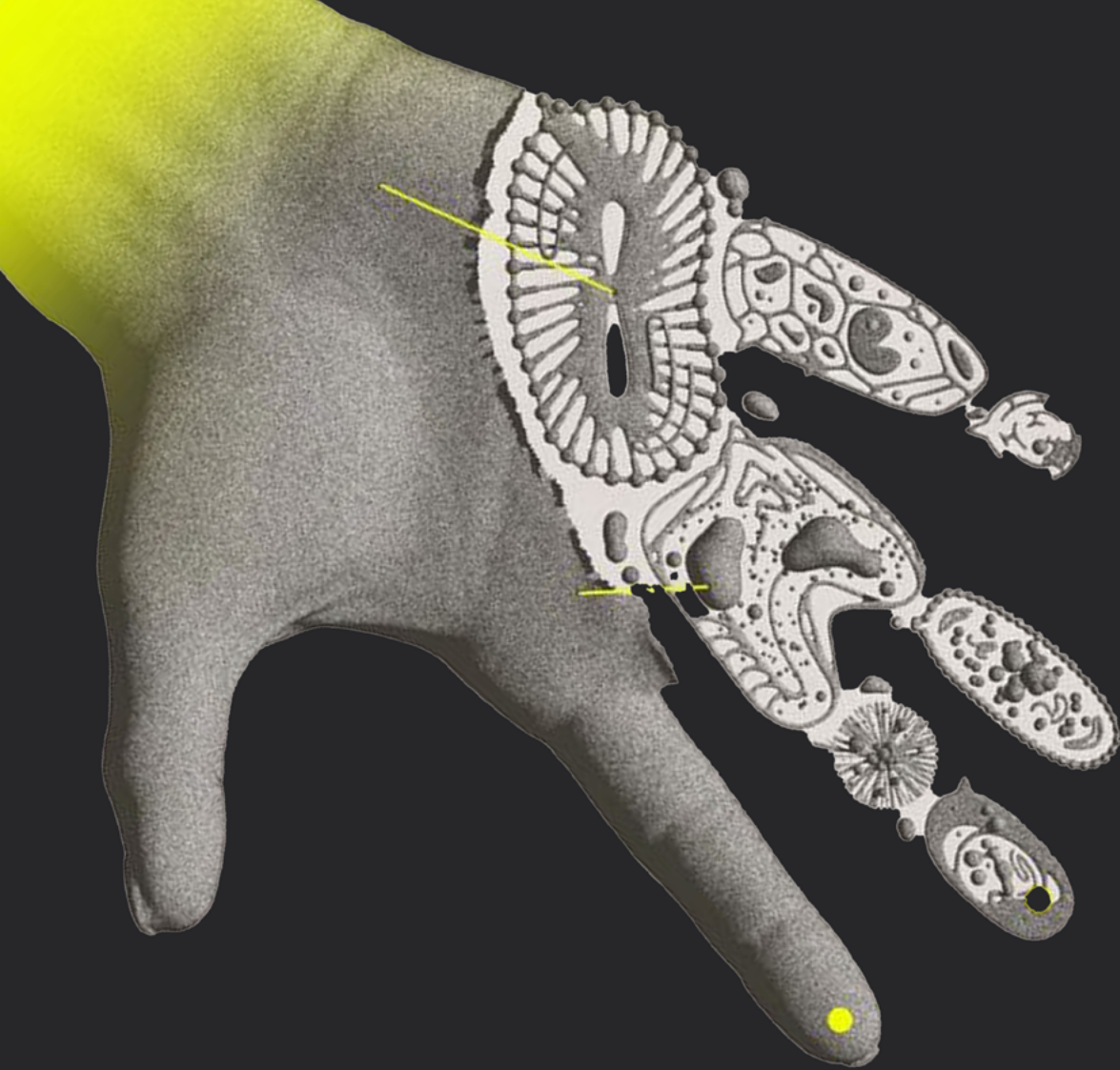
Traditional diabetic education relies mostly on textual materials, which are plentiful but rarely suited for children. Clinical psychologists have discovered that these passive kinds of learning might be difficult for children to understand.

In response to the limitations, my research focuses on leveraging the interactive capabilities of computer games to impart education through immersive and situational learning experiences. The urgency of this approach is underscored by the growing prevalence of Type 2 diabetes among children, a trend that poses severe health risks. Research indicates that younger individuals with Type 2 diabetes experience a higher rate of complications compared to their adult counterparts and children with Type 1 diabetes.

Previous studies have explored health-related games, primarily aimed at promoting physical activity and aiding stroke rehabilitation (Consolvo et al., 2006; Fujiki et al., 2008; Alankus et al., 2010); however, I found a gap in games specifically designed for children with Type 2 diabetes. The concept of "flow," first introduced by Csikszentmihalyi in 1990, is frequently used to explain the intense engagement observed in computer gaming. This paper goes into the conditions necessary for achieving engaged learning and explores how my findings lead to the identification of three key approaches for designing educational games.

For this design, I utilized an open-source variation of the iconic Mario Brothers game, "Mario Bros" (available at <https://github.com/mx0c/super-mario-python>), recognized for its ability to captivate children and adaptability across various computing platforms.

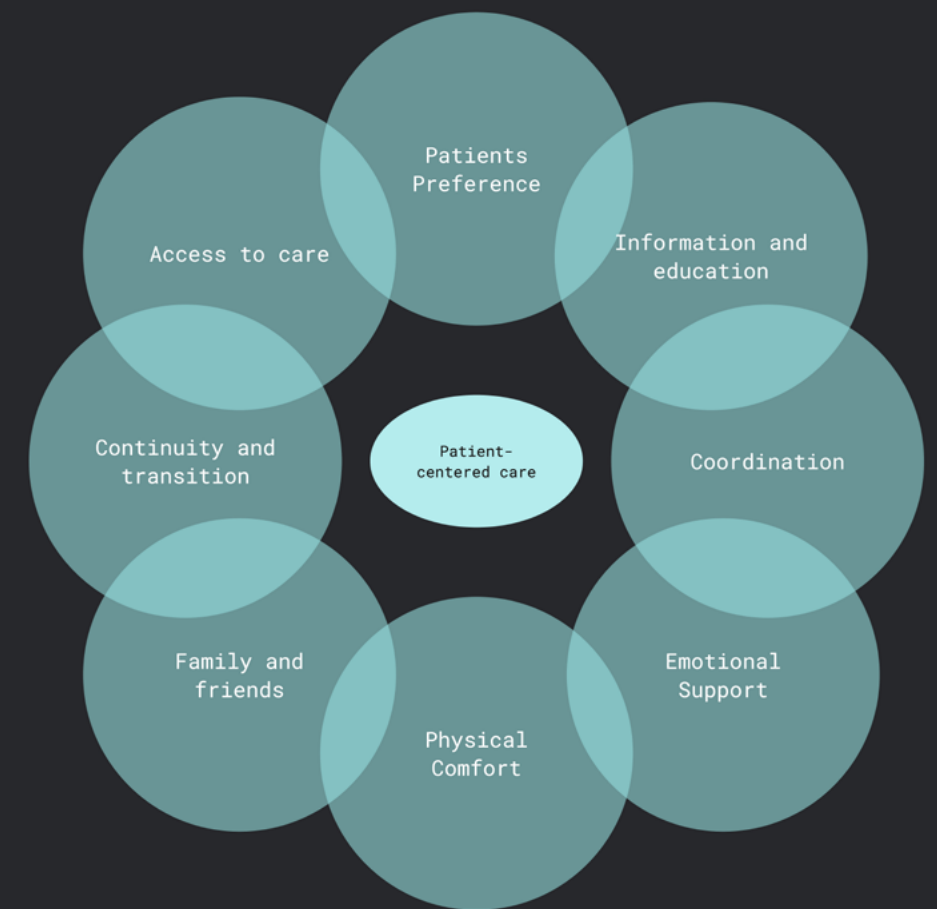
The document will discuss the key educational adjustments made to the game and report on an early assessment of its educational impact through tests conducted with children. The idea is to set a foundation for systematic design and development of comprehensive computer games focused on Type 2 diabetes education. The conclusion of the paper will outline potential avenues for future research.



Humanising Healthcare

It is significant that by the late twentieth century, medical and architectural literature began to recast traditional design features in terms of '(re)humanization'. This term sets a clear contrast against the modern hospital, frequently criticized for its impersonal nature and representation of contemporary medical practices.

Alongside the new technologies, there has been a growing concern about the loss of the "human" elements in medicine. (Bates 2018)



Introduced by Balint in 1969, patient-centered care has been developed as an alternative to the traditional, paternalistic, disease-centered model. Instead of focusing primarily on symptoms during the clinical encounter, the physician shows compassion and empathy towards the patient, respects her/his individual values, needs and preferences, and involves the patient in the decision-making process.

While patient-centered care is mainly visit- and episode-oriented person-focused care adopts a more holistic perspective, considering the patient as a person with a unique personal history, and treating symptoms and diseases in the context of the course of life.



"HEALTH AND HEALTHCARE TECHNOLOGY ARE
JUST SMALL PIECES THAT THEY [PATIENTS] TRY
TO FIT INTO THE LARGER PUZZLE OF THEIR
EVERYDAY ROUTINES"

- BALLEGAARD, HANSEN & KYNG 2008

Prevalence and IMPACT OF DIABETES

WHAT IS DIABETES?

Diabetes is a chronic disease that occurs either when the pancreas does not produce enough insulin or when the body cannot effectively use the insulin it produces.

537M

1 in 10 adults has
diabetes(20-79 yrs)(2021)

643M

By 2030, 1 adult in 9
will have diabetes

239.7M

Almost 1 in 2 adults with diabetes
were unaware of their diabetes
status

TYPE 1 DIABETES is an autoimmune condition in which the immune system is activated to destroy the cells in the pancreas which produce insulin.

We do not know what causes this autoimmune reaction. Type 1 diabetes is not linked to modifiable lifestyle factors.

There is no cure and it cannot be prevented.

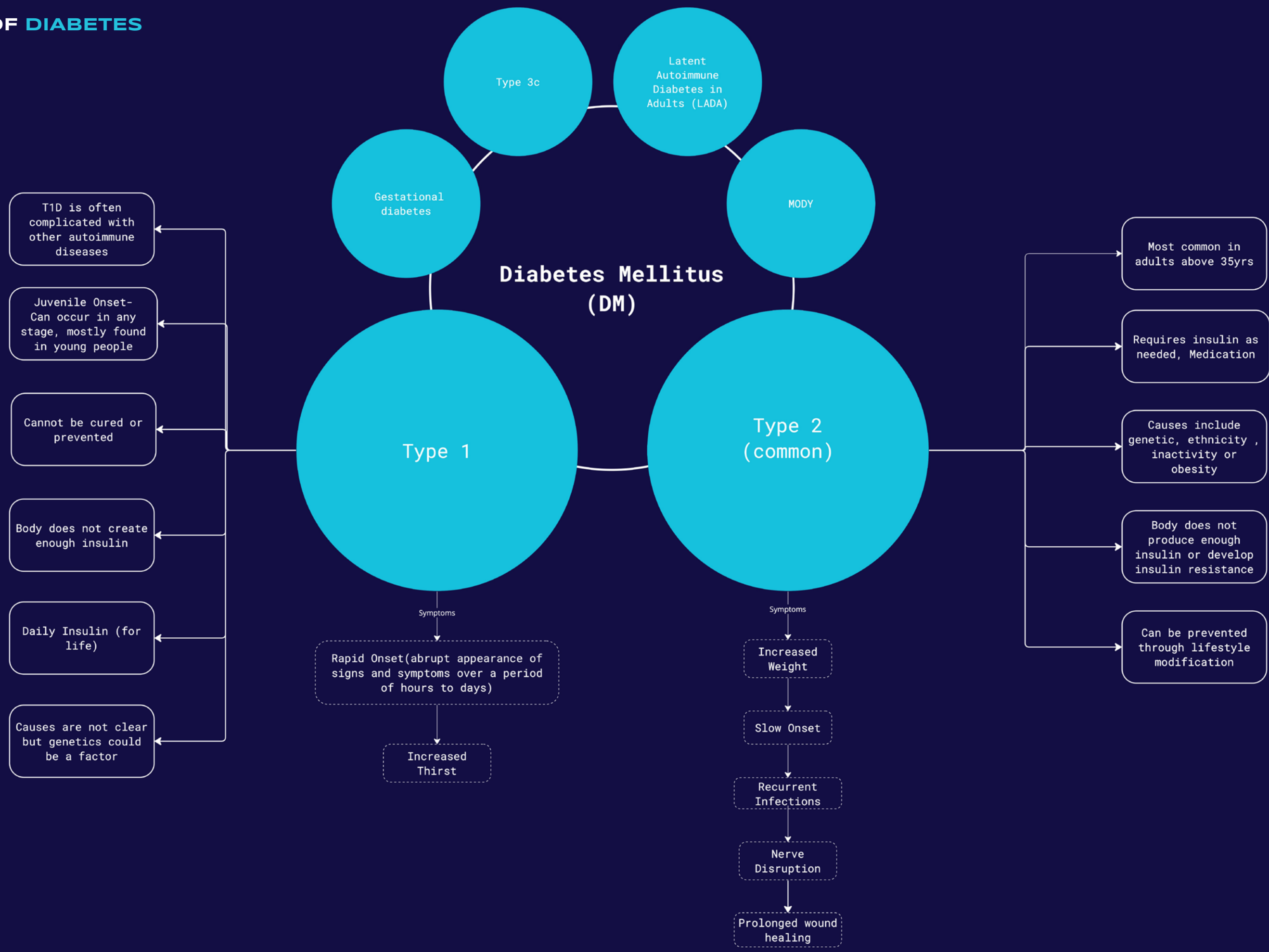
TYPE 2 DIABETES is a condition in which the body becomes resistant to the normal effects of insulin and gradually loses the capacity to produce enough insulin in the pancreas.

The condition has strong genetic and family-related (non-modifiable) risk factors and is also often associated with modifiable lifestyle risk factors. We do not know the exact genetic causes of type 2 diabetes.

People may be able to significantly slow or even halt the progression of the condition through changes to diet and increasing the amount of physical activity they do.

Evidence shows that around half of all people with recently diagnosed type 2 diabetes can achieve remission through intensive dietary changes and weight loss

TYPE OF DIABETES

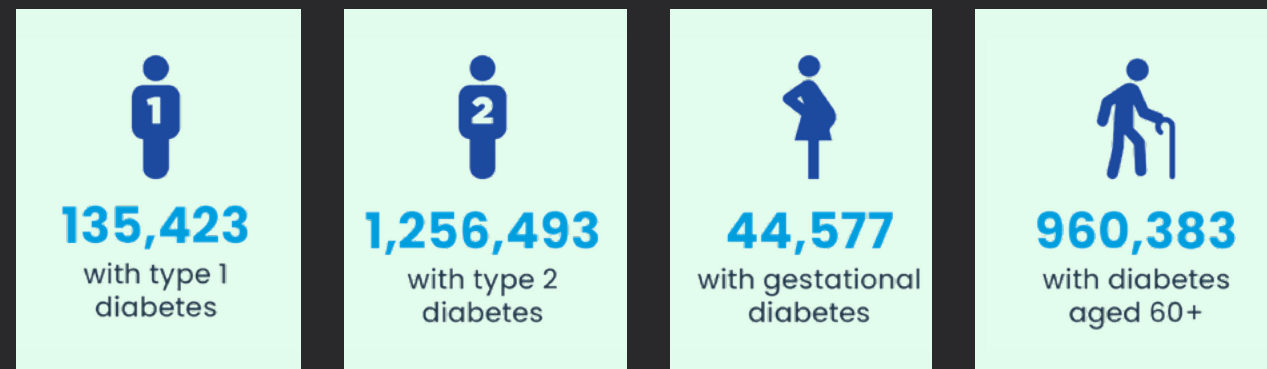


DIABETES IN AUSTRALIA

Diabetes is one of the most significant and complex public health challenge that Australia has faced.

Currently, around **1.5 million Australians**, or approximately **5.6%** of the population, are living with diagnosed diabetes and are registered with the National Diabetes Services Scheme (NDSS).

Statistics as of September 30th 2023



The past year has seen a significant rise in registrations, with **113,907 new cases of diabetes recorded**.

This equates to an alarming rate of approximately **312 new registrations each day**.

This data underscores the urgent need for increased awareness, better management strategies, and more effective prevention measures to combat this escalating health crisis.

PRE-DIABETES IN AUSTRALIA

An additional layer of concern is the prevalence of pre-diabetes, a condition indicating a high risk of progressing to type 2 diabetes

2M

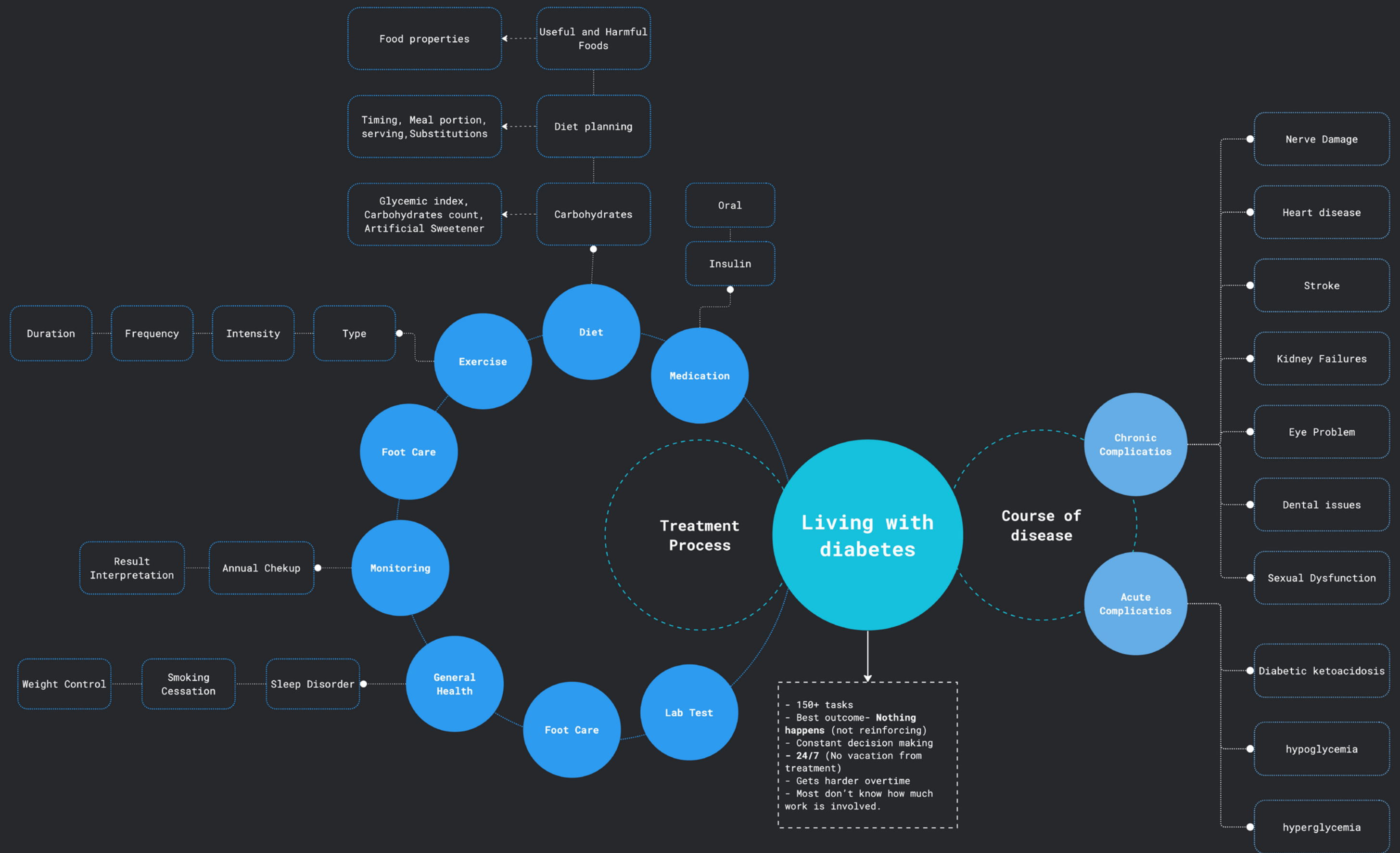
4M

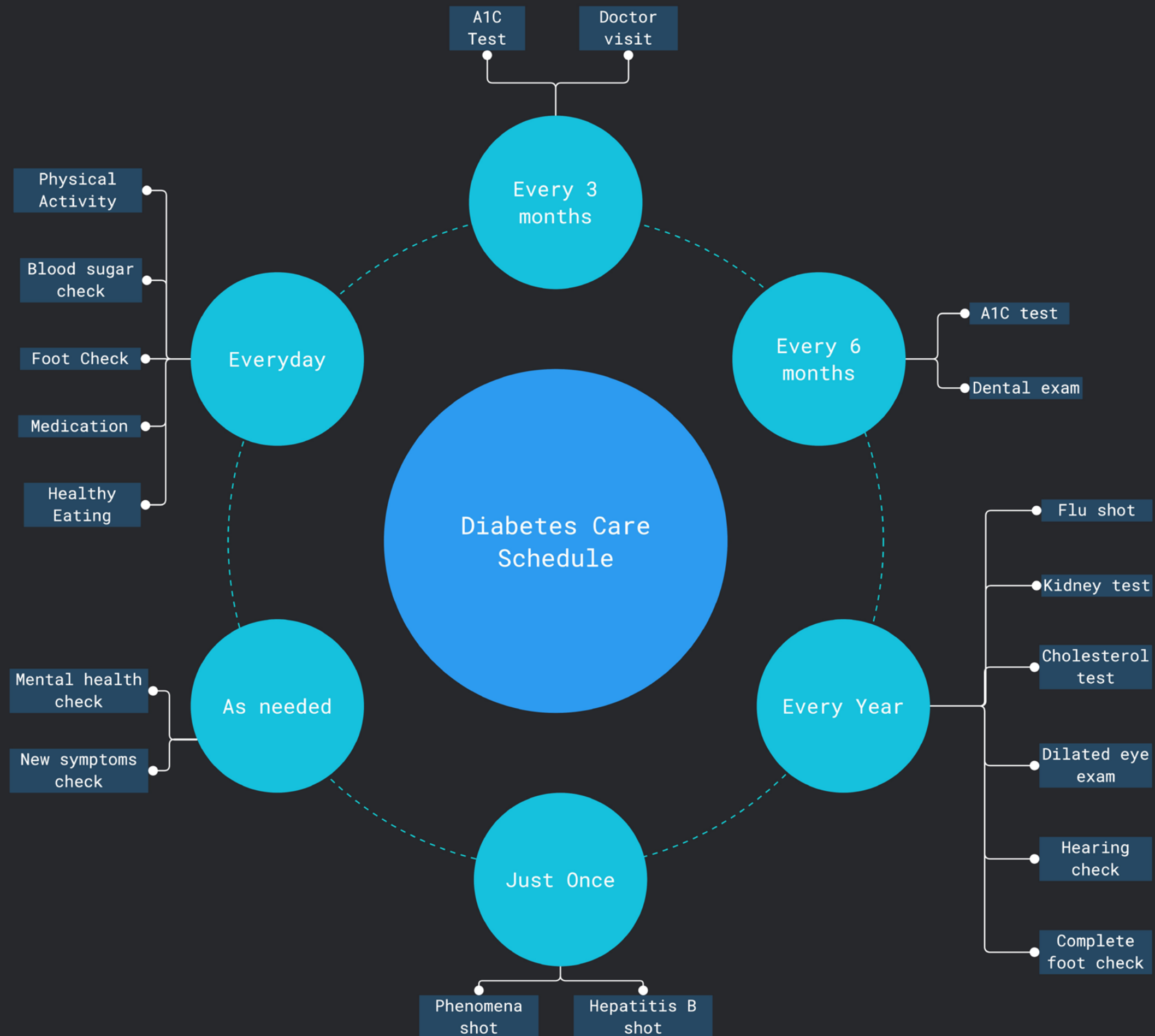
1 in 6 adults over the age of 25 are living with pre-diabetes

15% of the Australians, are either currently managing diabetes or are at significant risk of developing the condition.

A recent study reveals that the prevalence of type 2 diabetes has tripled in Australia from 1990 to 2019.

THERE IS AN URGENT NEED FOR COMPREHENSIVE STRATEGIES TO MANAGE AND MITIGATE DIABETES, ENCOMPASSING NOT ONLY THOSE CURRENTLY DIAGNOSED BUT ALSO THOSE AT IMMINENT RISK.





concerning trend in early onset of type 2 diabetes

Over the past two decades, there has been a significant increase in the incidence of Type 2 diabetes among younger populations, including children, adolescents, and young adults. This is largely attributed to rising levels of obesity (Zeitler et al., 2018; Misra et al., 2022).

Type 2 diabetes, once mainly seen in older adults, is increasingly affecting younger individuals in Australia and worldwide.

The definition of early-onset type 2 diabetes is generally applied to those under 40 years old, with further categorizations into children and adolescents (under 18 years) and young adults (18 years and older). Although these age divisions are somewhat arbitrary, they reflect a clear shift in the demographic affected by this disease (Misra et al., 2022; RACGP, 2022).

It is estimated that young adults now comprise 16% of the global adult population with type 2 diabetes (ADS, ADEA, & APEG, 2022).

Research indicates that younger individuals with type 2 diabetes experience a higher rate of complications compared to their adult counterparts and children with type 1 diabetes. Data from a multi-center observational study spanning 2011 to 2020, which involved 500 adolescents from the TODAY study, revealed that significant complications arose just 13 years after diagnosis, surpassing rates seen in pediatric type 1 diabetes patients and adults with type 2 diabetes (TODAY Study Group, 2020).

The ongoing trend of rising type 2 diabetes among younger populations is poised to have deep impacts both on individual health and on societal socioeconomic conditions. Therefore, early recognition and management of type 2 diabetes in younger populations are crucial to address this escalating problem.



(Goldman & Goldman 2022, fig. 2)



As we increasingly prioritize mental well-being, our pursuit of a better quality of life becomes more central

The World Health Organization defines Quality of Life (QOL) as an individual's perception of their position within the cultural and value systems they inhabit, and how this aligns with their aspirations, expectations, standards, and concerns.

The concept of quality of life is inherently complex and subjective, making it challenging to reach a consensus on its definition.

"THE QOL AMONG DIABETIC IS AN IMPORTANT PARAMETER FOR TREATMENT AND CAN AFFECT PATIENTS' METABOLISM"

(MORADPOUR ET AL. 2022)



(Posford 2023, fig. 5)

TYPE 2 DIABETES REDUCES PEOPLE'S QOL

- Social
- Physical
- Psychological well-being
- increase the economic cost to individuals and society

(Janssen et al. 2020)

QUALITY OF LIFE and HEALTH LOCUS OF CONTROL

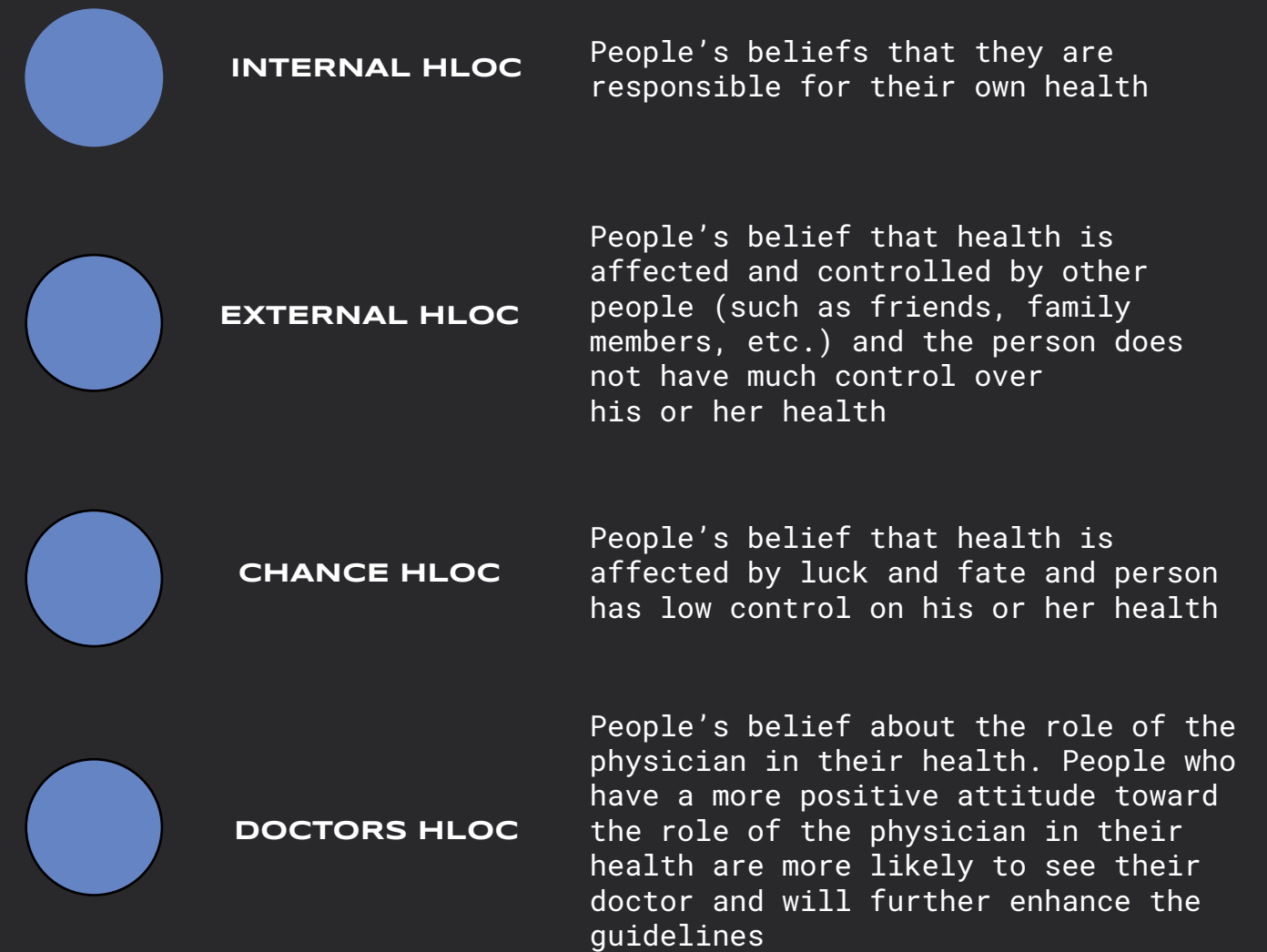
While significant efforts are being made to increase routine monitoring of psychological well-being and understand the lived experience, no guidelines currently recommend routine clinical assessment of diabetes-specific quality of life, and there is no consensus on which questionnaire(s) to use.

Type 2 diabetes is a complex and multifaceted nature and is affected by different factors. To prevent serious complications and death in people with diabetes, managing diabetes requires self-care behaviors in terms of choosing healthy foods, engaging in physical activity, taking appropriate medications, and controlling blood sugar.

" PRACTICING SELF-CARE BEHAVIORS IS ONE OF THE FACTORS THAT CAN HELP IMPROVE QOL OF PEOPLE WITH DIABETES."

One of the factors influencing self-care behavior is the **Health locus of control points (HLOC)**

HLOC is a psychological variable can predict diabetic self-management behaviors (Zhu et al. 2021)



The results of one study showed that the internal HLOC had the greatest influence on predicting self-care behaviors in patients with type 2 diabetes. There was also a significant relationship between the internal HLOC and regular medicine use among diabetic. (Zeidi, Morshedi & Otaghvar 2020)

One of the factors affecting the source of health control is health literacy.

Health literacy(HL) refers to “the extent of people’s ability to obtain, process and understand basic health information and access services needed to make appropriate health decisions”. (Marciano, Camerini & Schulz 2019).

People with diabetes need to become familiar with the scope and complications of the disease to manage their condition.

Studies have shown that health literacy increases health- promoting behaviors, reduces disease complications and improves QOL.

(Jafari et al. 2023)

ISSUES WITH LOW HEALTH LITERACY

- Lower engagement with health services (Kobayashi et al. 2014)
- Higher hospital re-admission rates (Mitchell et al. 2012)
- Poorer understanding of medication instructions (for example, non-adherence, improper usage) (Marvanova et al. 2011; Miller 2016)
- Lower ability to self-managed care (Geboers et al. 2016).

THE 2 MAIN COMPONENTS OF HEALTH LITERACY

- **Individual health literacy** - These are individual skills, such as the ability to find, understand and use health information; for example, to complete health care forms or understand and use the health care system.
- **The health literacy environment** - these are the health system-based elements, such as policies, processes, and materials, which affect the way the individual engages with the health system (ACSQHC 2014a).

HEALTH
Literacy

HEALTH LITERACY HAS THREE LEVELS

- **Basic or functional level** - skills to read and write in daily life
- **Communicative or Interactional level** - Social, cognitive as well as literacy skills
- **Critical level** - Able to use knowledge and deal with situations

HEALTH LITERACY AND DIABETES

Health literacy (HL) is crucial for effectively managing and understanding diabetes.

The World Health Organization (WHO) recognizes HL as a significant preventive factor against type 2 diabetes, associating low HL with modifiable risk factors such as smoking, poor diet, and physical inactivity. (Tajdar et al. 2021)

Research indicates that adults with inadequate HL are more likely to misjudge their risk of developing diabetes compared to those with adequate HL. This misjudgment is attributed to a general lack of awareness about chronic diseases among individuals with low HL.

As a result, those with low HL and a high risk of diabetes could greatly benefit from educational programs aimed at enhancing their understanding of diabetes.

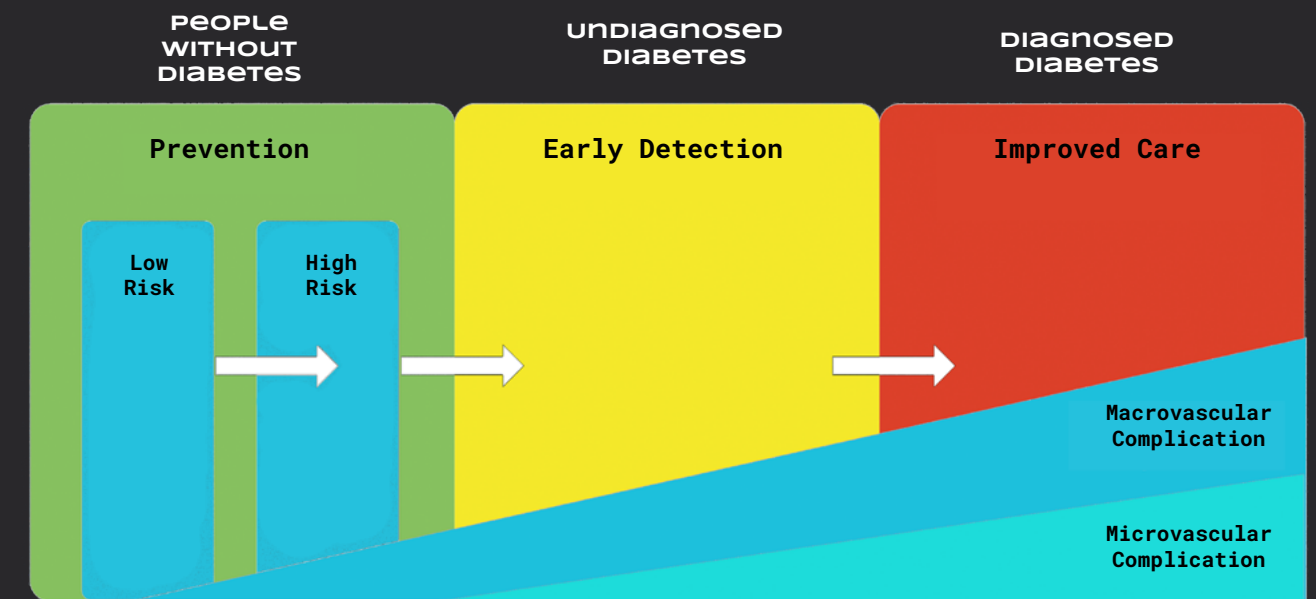


**Focus on
Prevention**

SUBSTANTIAL research CONFIRMS THAT LIFESTYLE MODIFICATIONS, SPECIFICALLY ENHANCING **PHYSICAL ACTIVITY and DIETARY HABITS** PLAY a crucial role in SIGNIFICANTLY REDUCING THE RISK OF DIABETES.

(CARDONA-MORRELL ET AL., 2010; SCHELLENBERG ET AL., 2013).

PREVENTING DIABETES BURDEN (DIABETES AUSTRALIA, 2013)



The macrovascular complications associated with diabetes, such as heart disease, stroke, and peripheral vascular disease, can commence even before the onset of diabetes. On the other hand, microvascular complications, including damage to the eyes, kidneys, and nerves, begin to manifest with the development of diabetes.

However, due to delayed diagnoses, many individuals already have these complications at the time of their diagnosis.

To mitigate the impact, a comprehensive strategy is essential.

This strategy should focus on preventing diabetes in both high-risk individuals and the general population.

There is substantial evidence supporting the effectiveness of these approaches in diabetes prevention, early diagnosis, and improved management.

EDUCATION FOR PREVENTION

Diabetes self-management through education is fundamental to diabetes therapy.

THERE IS A SIGNIFICANT
LACK OF LITERATURE AND
RESOURCES TAILORED TO
THOSE DIAGNOSED UNDER
THE AGE OF 30.

Educational programs designed for older adults may not meet the specific needs of younger individuals, and it is inappropriate to mix these age groups in educational settings.

The TODAY study indicates that traditional lifestyle interventions might be less effective for young adults with type 2 diabetes (YT2D), necessitating further investigation. Key barriers include time constraints, lack of motivation for rigorous self-management, and underestimating the seriousness of their condition, which often leads to poor follow-up.



("Trends in Diabetes Among Young People" 2024, fig. 7)

Interactive education tools, such as the Conversation Map, can foster effective interaction, peer support, and problem-solving skills.

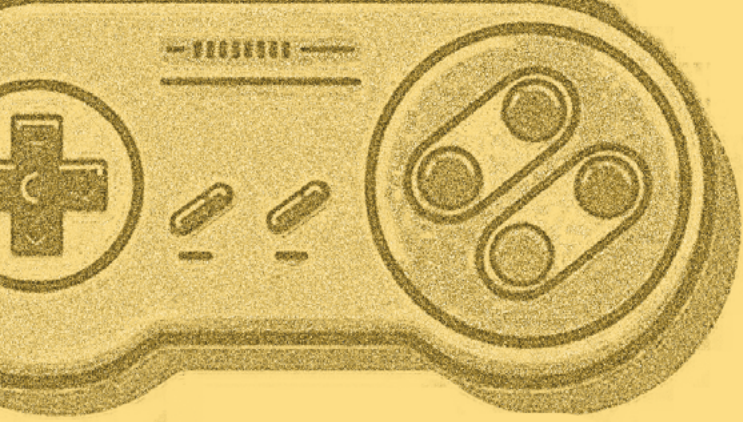
For some, visual expressions of their feelings about diabetes can be beneficial.

Written resources should be aimed at an 11-year-old reading level. **Additionally, apps that incorporate gamification, motivating users through competition with themselves or others, can enhance engagement by tracking, monitoring, and achieving health goals.**

(Bjornstad et al. 2021)



("Prediabetes and Type 2 Diabetes in Children and Adolescents: Screening" 2022, fig. 8)



GAMES SHOULD NOT BE TREATED SIMPLY AS EDUCATIONAL “SUGAR COATING”, MAKING THE HARD WORK OF LEARNING EASIER TO “SWALLOW”.

(RIEBER ET AL,1998)



There are five broadly defined prerequisites for effective learning:

- **Meaningful context**- A strong and engaging narrative
- **Authenticity** - Purposeful interactions (with feedback)
- **Motivation** - Emotional engagement
- **Freedom to make mistakes** - Freedom to fail
- **Learner autonomy**- Opportunities for independent decision making

Louisa Rosenheck, a researcher at MIT’s Education Arcade Lab, highlights the intrinsic motivation in effective learning games

“What’s really motivating about a good learning game is the learning,”

**VIDEO GAMES
AND EDUCATION**

Humans naturally seek to learn and improve, but Rosenheck points out that many educational games in classrooms focus too heavily on content and procedural skills, failing to spark genuine engagement and intrinsic motivation.

She advocates for “**resonant games**” which are open-ended, exploratory, and allow players to connect with broader systems and concepts for deeper learning.



Offer players choice and control over their gameplay and goals.



Spark curiosity, making players ask more questions and wonder how things work.



Provide “hard fun” – an appropriate level of challenge that is engaging and satisfying.

Truly educational video games prioritize authentic learning over simple “gamification,” which often relies on superficial rewards like points and badges. Such rewards can detract from the learning process by attempting to “trick” kids into learning

Instead, educational video games prioritize authentic learning by:



Encourage independent exploration and problem-solving without heavy reliance on direct instructions.



Provide feedback that helps players understand game mechanics and apply them to learn new concepts.



Promote connections between the game content and real-world challenges and systems

Many adults often limit gaming time, but it’s important to recognize that fun and learning can go hand in hand. The real challenge is not just finding high-quality games, but also helping kids make connections between their gameplay and the real world.

"FLOW"

a concept first introduced by Csikszentmihalyi in 1990, is often used to explain the level of engagement in a computer game.

Malone (1980) suggests that several conditions are likely to induce a state of flow. Among them, a few conditions are particularly important for designing diabetes education games:

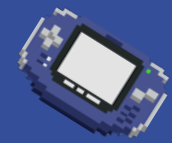
Challenge Enhancement (CE): Gradually increase the difficulty of educational content through stages. Start with simple concepts and advance to complex ones to stimulate proactive learning.

Feedback Enhancement (FE): Provide immediate, knowledge-rich visual feedback based on the game character's actions. Use pop-ups, message boxes, and performance indicators to reinforce learning.

Structure Enhancement (SE): Integrate educational features without disrupting the game's structure. Ensure that learning is a seamless part of the gaming experience, enhancing engagement through progressive challenges and rewards.

Malone's principles suggest that instead of disguising educational elements as fun activities, it's more effective to thoughtfully design the game's structure to promote engagement and active learning.

Project
Ideation



HOW MIGHT we effectively integrate diabetes education into video games to engage children for immersive learning?



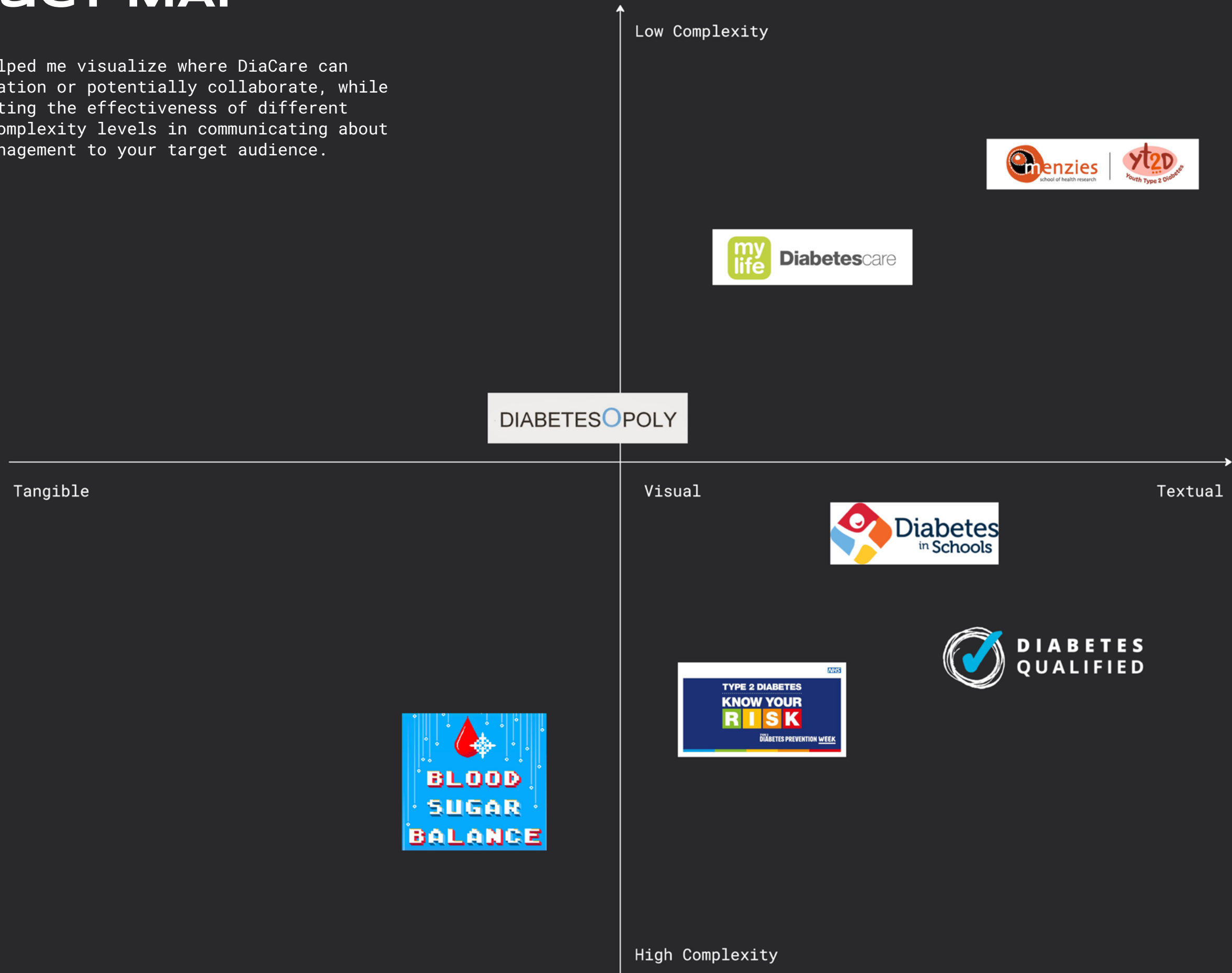
Goal

The primary objective of this project is to improve and encourage discussions about blood sugar management in Type 2 diabetes among children and teenagers.



Impact Map

This map helped me visualize where DiaCare can draw inspiration or potentially collaborate, while also indicating the effectiveness of different media and complexity levels in communicating about diabetes management to your target audience.





Diabetes Life Course (Brochure)

Offers straightforward, easily accessible information on diabetes management and health services in Central Australia, tailored for local communities.

Complexity: Low Complexity
Media Type: Textual



My Life Diabetes Care

Offers resources and support systems for individuals managing diabetes, with a focus on personal diabetes management through technological aids.

Complexity: Low Complexity
Media Type: Visual



Diabetes UK

Focuses on raising awareness and providing preventative tips and strategies through visually engaging materials. The initiative targets a wide audience, emphasizing the importance of early action.

Complexity: High Complexity
Media Type: Visual



Diabetesopoly Game

A board game designed to educate players about diabetes management in a fun and interactive way, suitable for all age ranges.

Complexity: Low Complexity
Media Type: Tangible, Visual



Diabetes in Schools

A comprehensive resource providing detailed guidance and tools for managing diabetes in a school setting, including educational materials for both staff and students.

Complexity: High Complexity
Media Type: Textual, Visual



Diabetes Qualified

Provides in-depth, structured education on diabetes care, specifically designed for childcare workers to ensure safe and informed care environments for children with diabetes.

Complexity: High Complexity
Media Type: Textual, Visual

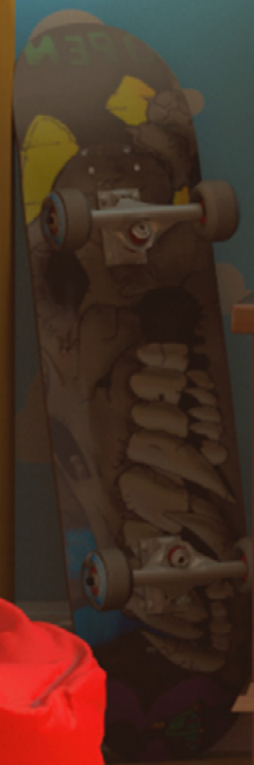


Blood Sugar Balance Game (NIH SEPA)

An interactive game designed to educate about glucose homeostasis through gameplay, targeting educational institutions and health educators to facilitate learning about the biological and lifestyle factors influencing diabetes.

Complexity: High Complexity
Media Type: Visual

DESIGN ~ game



Among all types of games,
simulation and role play games
are most likely to satisfy the
3 enhancement criteria's.

Learning by doing, through simulations and role-playing, is more effective and enjoyable than just studying facts.



WHY Mario?

After evaluating various open-source games such as Minetest, a version of Minecraft (benkyd/Minecraft.git), and Tetris (brenns10/tetris.git), and considering the development of a game from scratch, I ultimately chose Mario due to its strong alignment with the three conditions outlined in the Flow theory.

This decision was made after assessing the educational potential of each option, where Mario stood out as the most suitable platform for integrating the educational content effectively.

Main reasons of my choice include:

Engagement



The Mario Brothers game offers varied levels of difficulty, engaging children effectively

Accessibility



Being open-source and available in Java and Python, the game can be modified and adapted to various platforms increasing access to health education.

Interactive Feedback



Built-in feedback mechanisms encourage children to experiment with different strategies

Educational Integration



Stage-based design offers varied challenges, making it effective for embedding progressively complex diabetes knowledge at different stages of gameplay

Familiarity



Its long-standing popularity ensures a short learning curve and potentially broad acceptance among children and parents

Design considerations

Guided by three core design strategies, I have implemented educational features into the Mario Brothers game, modifying it to better suit the educational objectives.

In this adaptation, the protagonist, Mario, discovers that his future self is at risk of developing diabetes due to unhealthy habits.

In line with the fundamental principles of role-play game, Mario's health-related decisions are made into challenges that the player must navigate, thereby embedding educational content directly into the gameplay.

ICONS

Healthy Carbs



Apple



Broccoli



Grapes

Un-Healthy Carbs



Burger



Soda Can

Boost



Exercise



Insulin

To effectively manage his condition, Mario must monitor his Blood Glucose and Insulin levels across various stages of the game.

Early in the game's development, I introduced several icons to differentiate between healthy carbs, unhealthy carbs, and boosts:

Unhealthy Carbs: Items like Burgers and Coke, containing refined sugars, cause a rapid increase in blood sugar levels.

Healthy Carbs: Whole foods such as Fruits and Vegetables lead to a slower, more controlled rise in blood sugar.

Insulin Boost: This allows players to administer insulin to Mario, lowering his blood sugar when it becomes too high.

Exercise Boost: Beyond hormonal regulation, players can use exercise to decrease blood sugar levels quickly.

coding

The image shows a code editor with two files open: `main.py` and `drawBG.py`.

main.py (Left Panel):

```

1  import pygame
2  from classes.Dashboard import Dashboard
3  from classes.Level import Level
4  from classes.Menu import Menu
5  from classes.Sound import Sound
6  from entities.Mario import Mario
7  from config import SCALED_WIDTH, SCALED_HEIGHT
8  import sys
9
10
11 # Global flag to determine if the loading screen should be
12 first_run = True
13
14 def drawBG(screen, level):
15
16     mariox=2*32
17     marioy=12*32
18
19     koopax=17.5*32
20     koopay=11*32
21
22     # Drawing the background similar to the drawMenuBackgr
23     for y in range(0, 13):
24         for x in range(0, 20):
25             screen.blit(level.sprites.spriteCollection.get
26
27     for y in range(13, 15):
28         for x in range(0, 20):
29             if x==5 or x==6 or x==9 or x==10:
30                 screen.blit(level.sprites.spriteCollection
31             else:
32                 screen.blit(level.sprites.spriteCollection
33
34     screen.blit(level.sprites.spriteCollection.get("mario_
35
36     screen.blit(
37         level.sprites.spriteCollection.get("cloud1_1").ima
38     )
39     screen.blit(
40         level.sprites.spriteCollection.get("cloud1_2").ima
41     )
42     screen.blit(
43         level.sprites.spriteCollection.get("cloud1_3").ima
44     )

```

drawBG.py (Right Panel):

```

1  def main():
2      global first_run
3      pygame.mixer.pre_init(44100, -16, 2, 4096)
4      pygame.init()
5      screen = pygame.display.set_mode((SCALED_WIDTH, SCALED_HEIGHT))
6      sound = Sound()
7      dashboard = Dashboard("./img/fort.png./img/fort-red.png")
8
9      level = Level(screen, sound, dashboard)
10
11     if first_run:
12         show_loading_screen(screen, SCALED_WIDTH, SCALED_HEIGHT)
13         first_run = False # Set the flag to False after the first run
14
15     max_frame_rate = 60
16
17     menu = Menu(screen, dashboard, level, sound)
18
19     while not menu.start:
20         menu.update()
21
22     mario = Mario(0, 0, level, screen, dashboard, sound)
23     clock = pygame.time.Clock()
24
25     while not mario.restart:
26         pygame.display.set_caption(f"Super Mario running w
27         # if mario.speed:
28         #     print("SSSLL00000000wwhwwhww")
29         #     max_frame_rate = 30
30         if mario.pause:
31             mario.pauseObj.update()
32         else:
33             level.drawLevel(mario.camera)
34             dashboard.update()
35             mario.update()
36             pygame.display.update()
37             clock.tick(max_frame_rate)
38         return 'restart'
39
40 if __name__ == "__main__":
41     exitmessage = 'restart'
42     while exitmessage == 'restart':
43         exitmessage = main()

```

```
main.py [orchestrates the primary operations and interactions
of the game components]
```

Here I am importing necessary libraries and modules like `[pygame]` for game development, and specific game components such as `[Dashboard]` `[Level]` `[Menu]` `[Sound]` and `[Mario]`.

`[first_run]` is a global variable used to check if the game is being loaded for the first time to display a loading screen

```
main() :
```

- This function initializes the main game settings such as the display size, sound settings, and initializes core game components
- It checks if the game is running for the first time.
- It manages the main game loop, updating the menu and game level, drawing the level, updating the dashboard, and handling Mario's interactions.
- It also ensures the game runs at a specified frame rate.
- The game loop continues until Mario triggers a restart condition, at which point the game can restart or exit based on the return value from the main function.

```
Mario.py [Runs the behavior and functionalities of the main character - Mario]
```

Key Functions and Behaviors

- Update Cycle (**update**): This function is called repeatedly to handle Mario's behavior frame by frame. It checks for invincibility, updates traits, moves Mario based on velocity, applies gravity, checks for collisions, and processes input.
- Movement (**moveMario**): Adjusts Mario's position based on his current velocity and checks for collisions in both the horizontal (X) and vertical (Y) directions.
- Collision Handling (**checkEntityCollision**): This checks for and handles collisions with different types of entities like items, blocks, mobs, and special game-ending objects. Different actions are triggered based on the type of entity Mario collides with.

COLLISION RESPONSES

- Items (`_onCollisionWithItem`): When colliding with items (like power-ups), increase to glucose levels, and corresponding sound effects are played.
- End Level (`_onCollisionWithEnd`): This triggers the end of the level, potentially displaying different images based on the player's score, and setting up a restart.
- Unhealthy Items (`_onCollisionWithUnhealthy`): Colliding with unhealthy items results in a drastic increase to glucose levels.
- Blocks (Boost) (`_onCollisionWithBlock`): Interactions with blocks can trigger them, increase to glucose and Insulin levels and changing the block's state.
- Mobs (`_onCollisionWithMob`): Handles collisions with unhealthy items. Different reactions occur depending on the part of the mob Mario contacts (e.g., jumping on top vs. colliding from the side).
- Power-ups (`powerup`): Changes Mario's state, such as increasing size with insulin (blocks) intake or granting invincibility, triggered by specific items.



MARIO STANDING

COLLIDING WITH ITEM



COLLIDING WITH MOB

COLLIDING WITH BLOCKS



The diagram provides an outline of the behavior of Mario in the game, showing the effects of various actions and collisions with different elements.

The base code's effects are shown in red, while the updated effects for the "DiaCare" game are shown in green.

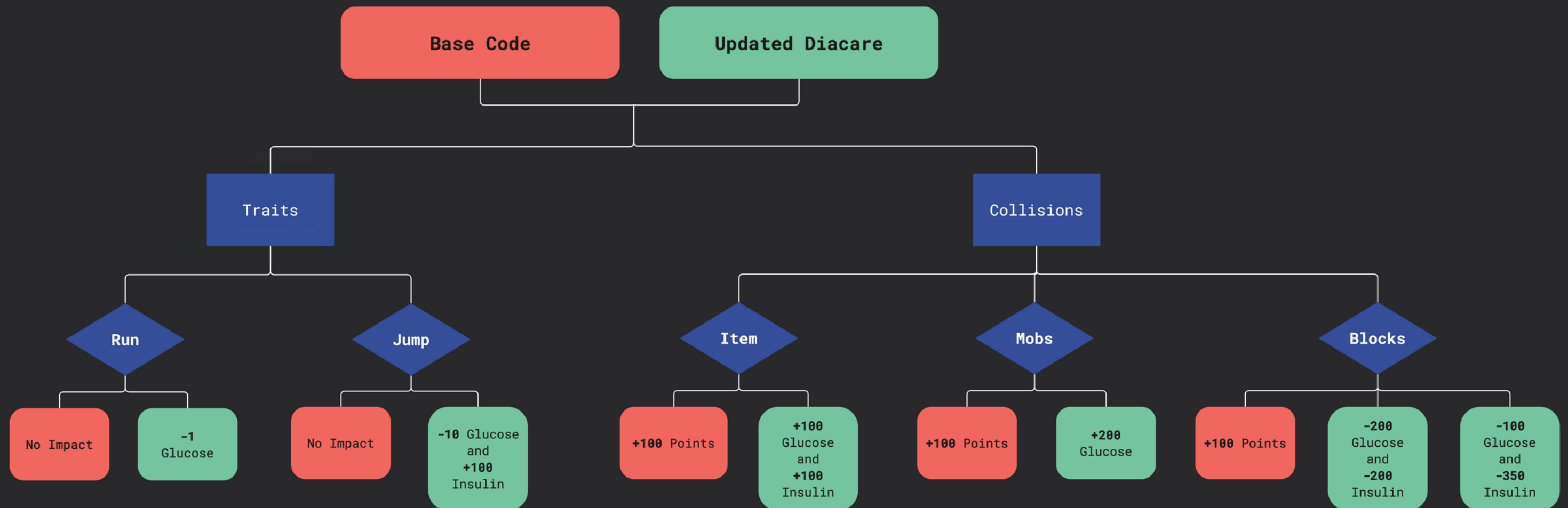
In the updated DiaCare game, changes focus on how physical activities and interactions impact health:

Running and Jumping: These activities now affect blood sugar and insulin, teaching players about the real-life effects of exercise.

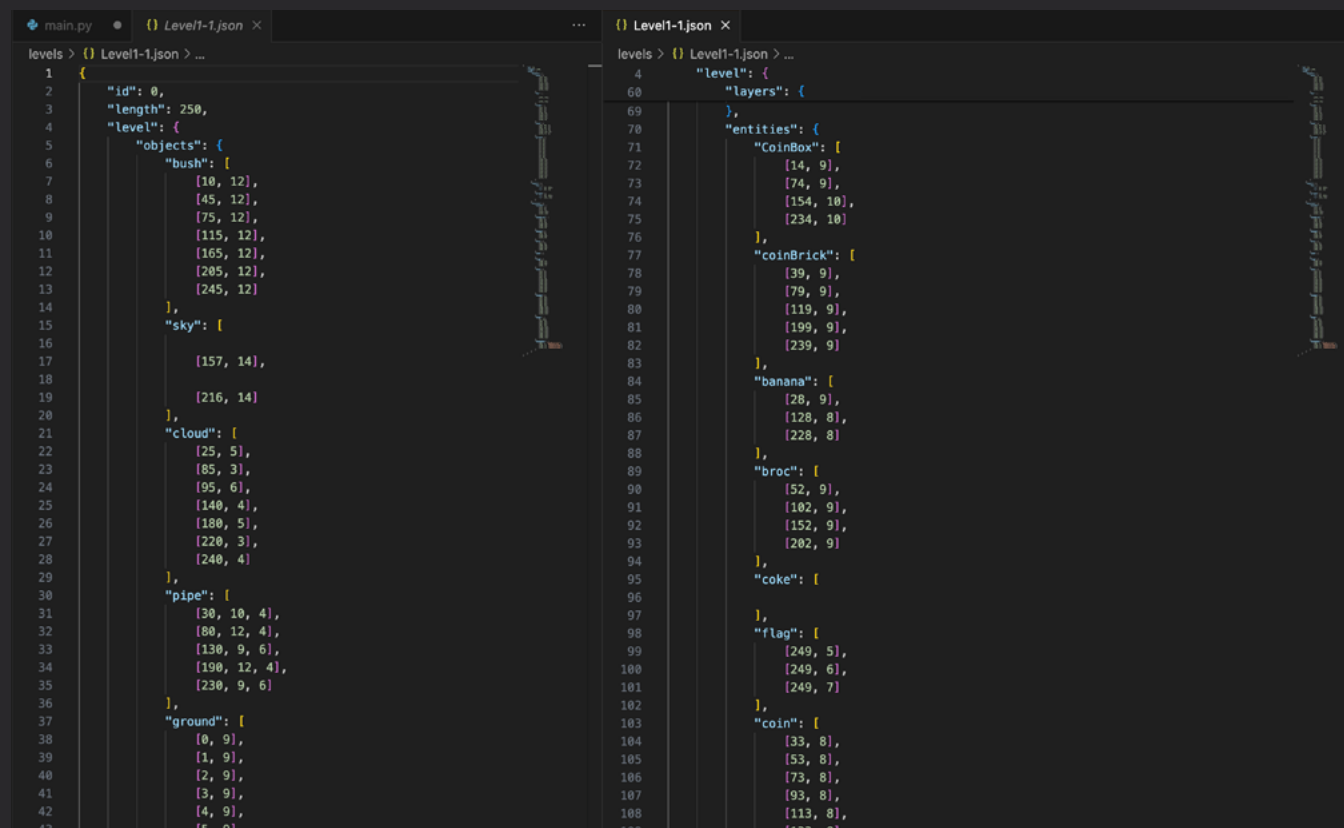
Items: Picking up items is good for health, improving health indicators.

Mobs and Blocks: Colliding with these is more challenging, having various effects on health metrics, which adds complexity to the game.

CODE FLOWCHART



Level overview



The **JSON file** outlines the structure and elements for Level 1 of the game (Blood Sugar Management)

There are 2 other JSON file for the Pre-Diabetic and Diabetic levels.

Level Length: The current level spans 250 units, represented in game tiles. This length can be extended as I further develop the game.

Objects and Their Placement

Objects are various elements placed throughout the level at specific coordinates, defined as [x, y], where x is the horizontal position and y is the vertical position on the game map.

- **Bushes:** Positioned at various points along the level to enhance the environment aesthetics.
- **Sky and Clouds:** Decorative elements to add to the level's atmosphere
- **Pipes:** Serve as obstacles or interactive elements, varying in height, placed at strategic points.
- **Ground:** Defines the walkable surface of the level. Continuous from the start to end of the level.

Entities

Box and Brick: Blocks that likely give boost when hit. Positioned throughout the level.

Fruits (Banana and Broc): Positioned to be collected by the player, affecting game stats.

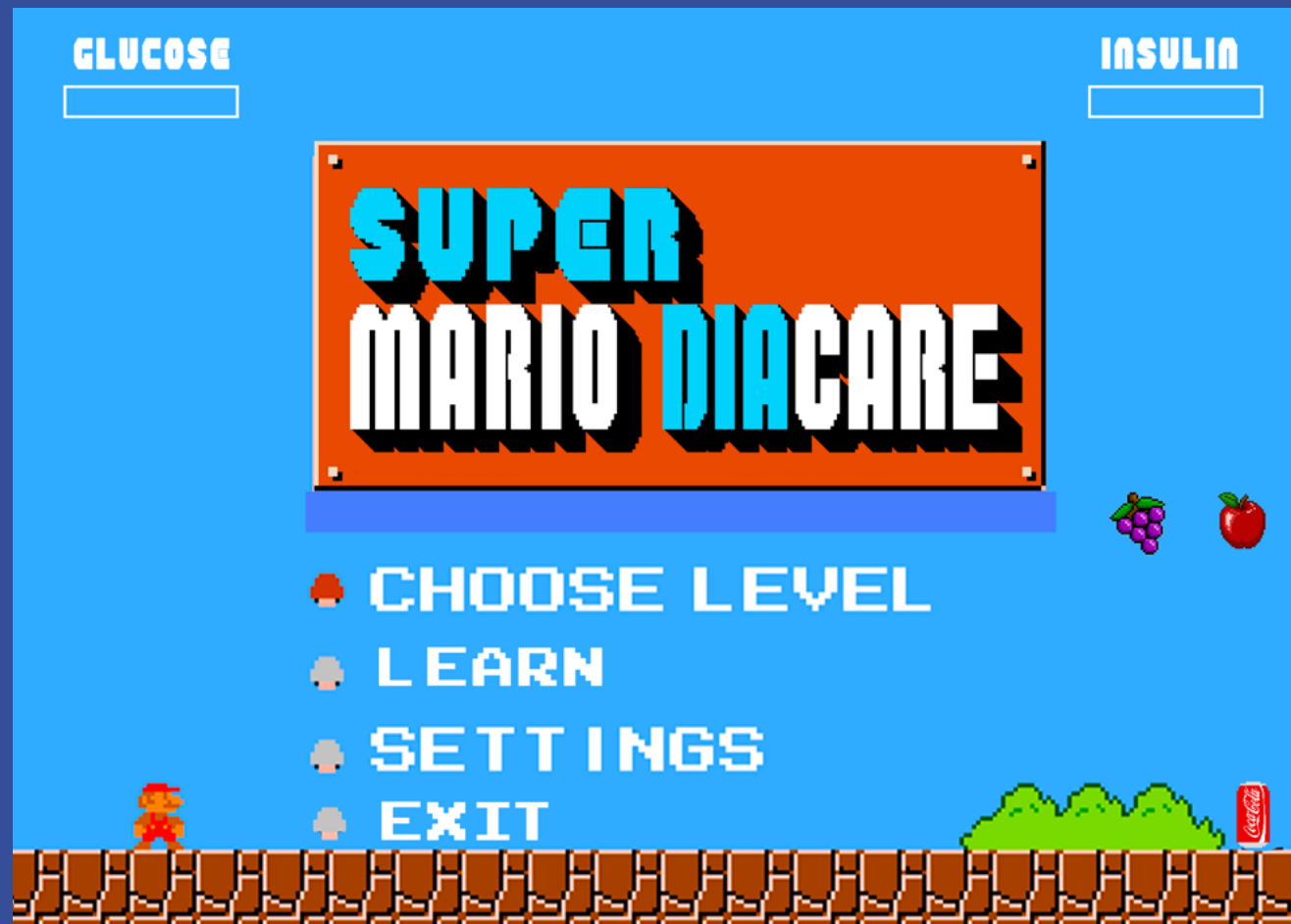
Coke and Burger: Positioned to be collected by the player, affecting game stats.

Flag: Marks the end of the level or checkpoints.

Game

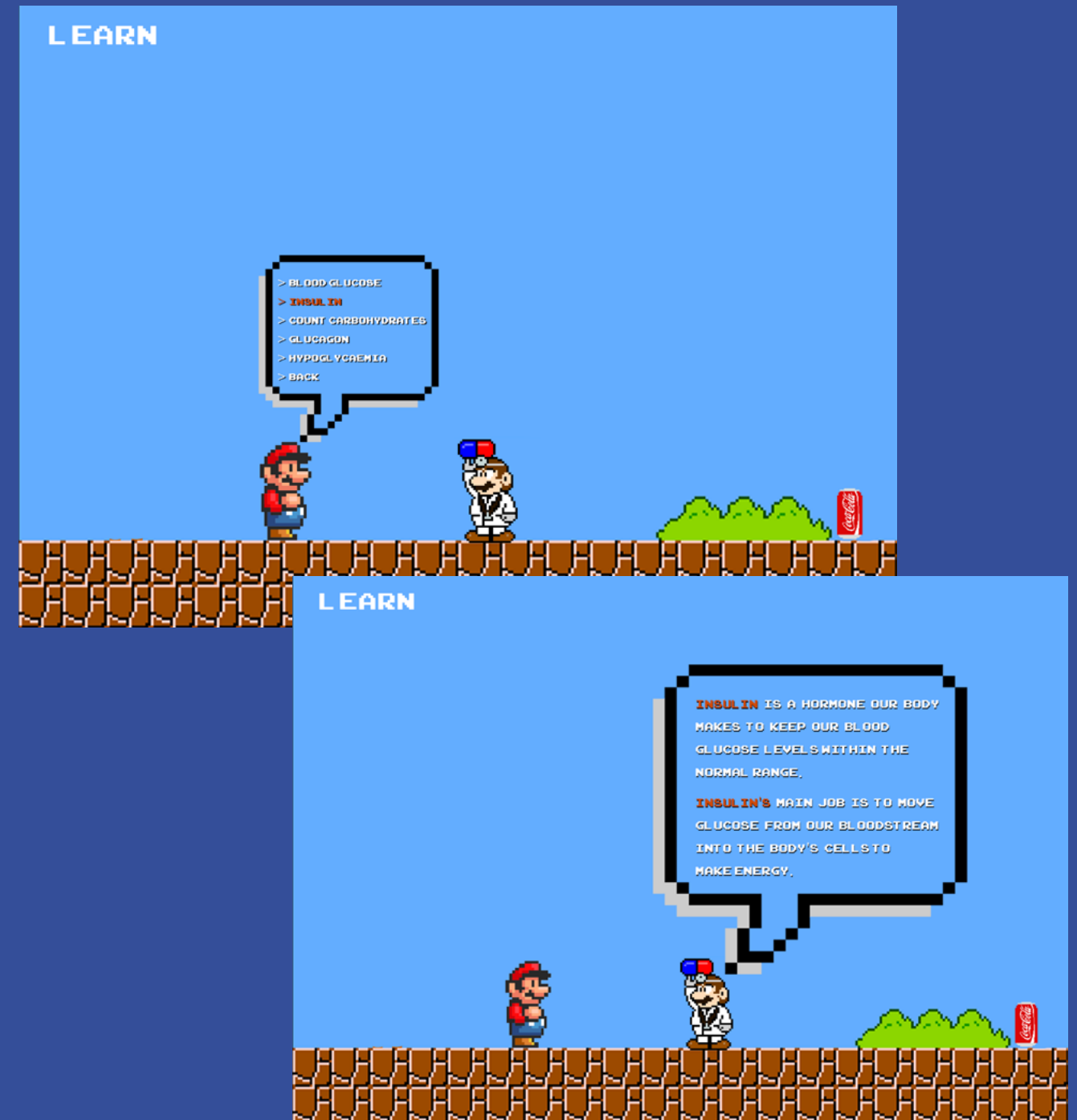
DiaCare offers an innovative approach to help kids and teenagers understand and manage Type 2 Diabetes by gamifying education and promoting proactive lifestyle changes.

DASHBOARD



Mario discovers that his future self is at risk of developing diabetes due to unhealthy habits, and he must take action to change his lifestyle and avoid this fate.

Learn



Mario meets a friendly character, Dr. Mario, who explains key terms related to diabetes.

Through engaging dialogue lessons, players gain a foundational understanding of diabetes, preparing them for the challenges ahead in the game.

Levels



The level of difficulty is adjusted to match the children's knowledge of diabetes, ensuring an engaging and educational experience tailored to their understanding.

Level 1 (BLOOD sugar management)



The players must help Mario maintain his blood sugar levels within a healthy range to rescue his future self.

Level Guidance: As the level begins, informational pop-ups appear on the screen. These provide concise tips and essential information about gameplay mechanics, helping players understand how their actions influence Mario's health.

Indicator Bars: Displayed at the top of the screen are the Blood Sugar and Insulin Bars, showing Mario's current levels. The bars change color to provide immediate feedback when levels reach, exceed, or fall below the ideal glucose range of 40-60%.

Insulin Dynamics: Although Mario is not diabetic, the game models a basic insulin response to dietary intake and physical activities, illustrating concepts of energy balance and metabolic health.

Food Item Collection: Players navigate Mario through the level to collect various food items, each affecting the Blood Sugar Bar uniquely:

- Apples and Bananas: Gradually increase blood sugar levels.
- Coke Cans and Burgers: Cause a rapid spike in blood sugar levels.

Physical Activity: Players are encouraged to have Mario engage in activities like jumping and running, which subtly lower his blood sugar level, mimicking the impact of exercise on metabolism.

BOOSTS:

Insulin Boost: Helps Mario utilize his current blood glucose more effectively, transferring it to his organs. This not only increases his energy but also causes him to appear larger, providing visual feedback.

Exercise Boost: Stimulates Mario's insulin production and enhances glucose conversion, boosting his energy similarly.

<https://github.com/ashikacode/DiaCare-Mario->

Game Play



<https://vimeo.com/955784249?share=copy>

TESTING



Ashika Ramesh, fig 10

Participant Ages

Tested the game with children aged 10 and 13 years

Initial Reaction

Curios about the game's characters and visuals



Ashika Ramesh, fig 11

Understanding Game Logic

Encountered some challenges in understanding the game's mechanics initially.

Once they understood the objectives, their engagement increased.

Educational Impact

Sparked interest about blood sugar and insulin

INTRODUCE DETAILED FEEDBACK MECHANISMS to guide player choices regarding food items, enhancing educational outcomes.

DEVELOP ADDITIONAL LEVELS to cater to older children, increasing the game's appeal and challenge.

INTEGRATION OF REAL-WORLD DATA to explore the feasibility of incorporating actual patient data to simulate real-life scenarios, enhancing the game's educational value.

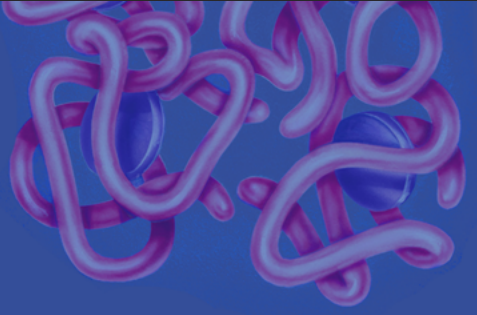
IMPLEMENT DYNAMIC CONSEQUENCES in the game, such as the character experiencing health crises (e.g., going unconscious) if blood sugar levels are poorly managed.

EDUCATIONAL INTEGRATION Plan to integrate "DiaCare" into preschool and middle school curricula as part of health education.

DEVELOP AND REFINE AN ACTIVITY KIT to be used post-gameplay, designed to facilitate further discussion and engagement on diabetes management and healthy lifestyle choices.

FUTURE WORK

ACTIVITY KIT (IDEATION)



Carbohydrates

Like proteins and fats, carbohydrates are one of the three main nutrients in food. Carbohydrates (carbs) are the body's major source of energy.

The two main types of carbs are:

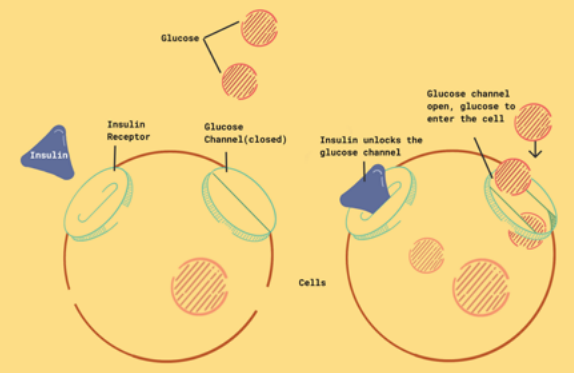
- **sugars**, like the kinds in milk, fruit, table sugar, and candy
- **starches**, found in grains, breads, crackers, and pasta

Sugars- Sugars are easily and quickly digested because they do not need to break down much, if at all, before entering the bloodstream.

Starches- Starches have bonds between glucose molecules that humans can break easily with digestive enzymes. Starches are polysaccharides.

Fibers- Fibers have bonds between glucose molecules that humans cannot break easily with digestive enzymes. Fibers are polysaccharides.

How Insulin Works?



After we eat, much of our food is broken down into **glucose (sugar)**.

Glucose travels in the blood to our cells where it is **used for energy** or **stored**.

With **type 2 diabetes**, too much glucose stays in the blood because it cannot get into the cell.

The system that maintains blood glucose balance is failing. **Insulin**, a hormone, is a key part of this system.

This causes chronic high blood glucose levels.

Over a period of time, high blood glucose levels can damage nerves, the heart, blood vessels, hands, feet, and kidneys.

Complications can even lead to death.

Supplementary Activities: To foster deeper discussion and application of the concepts learned, the kit also offers activities such as My Daily Plate and SMART Goal.

These exercises encourage children to apply their understanding in practical, everyday contexts, promoting continuous learning and engagement on diabetes management.

Educational cards- designed to enhance understanding of key diabetes-related concepts post-gameplay.

It includes diagrams and explanations of important topics such as carbohydrates and insulin.

These visual aids aim to solidify the knowledge acquired during the game.

Imagine how you might want to eat when you're older, and think about the good eating habits you'd like to share with your family or friends in the future.

Set one SMART goal right now that could help you keep eating healthy for a long time!

Specific	Measurable	Attainable	Relevant	Timely
S What is the goal?	M How will you measure your progress?	A Can you make this come true?	R Are you willing and able to do this?	T By when will you accomplish this goal?

My Daily Plate

Objective: To help the students become more aware of their daily eating habits by documenting and reflecting on their most common meals.

Materials Needed:

1. Paper plates (one per student)
2. Markers, colored pencils, or crayons

Activity Setup (10 minutes):

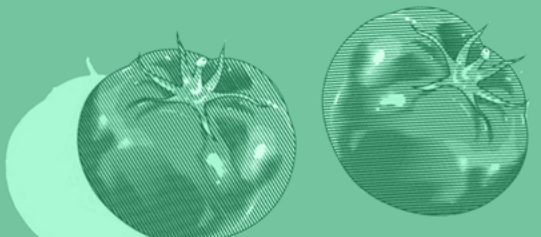
1. Give each student a paper plate and a set of markers or colored pencils.
2. The paper plate represents their meal plate for a typical day.

Meal Listing (15 minutes):

1. The students should draw and colour the foods they typically eat for breakfast, lunch, and dinner on the paper plate. They should divide the plate into sections for each meal.

Discussion and Reflection (15 minutes):

1. The students present their plates to the class and describe why these foods are part of their common meals.
2. Encourage them to discuss the nutritional value of these foods and discuss if their meal plate is balanced.



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